

Request for Proposal — Power, Earthing & Auxiliary Transformers

HESS Psevdas BESS — 63 MVA 132/33 kV step-up + 33 kV earthing/auxiliary transformer

Lighthief Cyprus Ltd (EPC contractor), on behalf of H.E.S.S. Hybrid Energy Storage Systems Ltd (the Client, project SPV), invites your firm offer for the supply of the transformers described below for the Client's standalone battery energy storage facility at Plot 26, Psevdas, Larnaca, Cyprus (TSOC connection ref. ΔΣΜΚ/ΠΟΣ/320.7.11).

HOW TO RESPOND: This RFP accompanies your transformer questionnaire, which Lighthief has pre-filled from the Client's requirements (the 'Required' column). Please complete the 'OFFER' column of that Excel for the main transformer (T1) and complete the attached earthing & auxiliary transformer datasheet (T2), and return both with a signed compliance statement, certificates, and your commercial offer.

1. SCOPE — TWO TRANSFORMERS

Quote T1 and T2 as SEPARATE line items. A bid covering only the main transformer is non-compliant. The earthing & auxiliary transformer (T2) is a separate unit — the TSOC connection terms list it as a distinct item — not a winding on the main transformer.

T1 — Main power transformer (63 MVA, 132/33 kV)

Quantity	1 unit
Rating	63 MVA continuous (ONAN / ONAF)
Voltage ratio	132 / 33 kV
Vector group	YNd11 (132 kV star / 33 kV delta)
Short-circuit impedance uk	21% on HV base @ 75 °C, CMR
Highest system voltage	145 / 36 kV
Insulation level — HV	550 kV BIL / 230 kV AC (1 min)
Insulation level — LV (36 kV class)	170 kV BIL / 70 kV AC (1 min) [corrects any 125/50 in legacy tables]
Neutral (HVN)	Power-frequency withstand 38 kV
Through-fault withstand	132 kV: 31.5 kA / 1 s 33 kV: 20 kA / 3 s
Cooling	ONAN / ONAF — 2 cooler banks @ 50% CMR each
OLTC	+12.5% / -18.75%, 25 × 1.25% steps; MR/ABB vacuum;

	local/remote/supervisory
Load power factor	0.9 lag (design)
Temperature rise	55 °C top oil / 60 °C winding
Sound level	≤ 60 dB(A) outdoor (≤ 80 worst case)
Pollution / creepage	Heavy — IEC 60815 Class III; composite ≥ 35 mm/kV
Standards	Tier 2 Ecodesign (EU 2019/1783); EN 60076 series; TSOC T14
Monitoring	Dual WTI (HV + LV hotspot); online DGA / transformer monitoring system
Oil / bushings / OLTC	EN 60296 oil; EN 60137 bushings; EN 60214 OLTC; silica-gel breather
Mounting / transport	Flat base; anti-vibration pads; EN 795 anchorage; impact recorders w/ GPS
Losses / PEI	To be stated by producer (no-load, load kW, PEI %)
Installation	Outdoor, oil-immersed; Plot 26 Psevdas, Larnaca, Cyprus

T2 — Earthing & auxiliary transformer (33 kV / 400-230 V)

Equipment	Separate earthing & auxiliary transformer (not a winding on T1)
Function	Derive & solidly earth the 33 kV system neutral + station-service supply
Earthing connection	Zig-zag (interconnected-star), ZN — SOLID earthing, no NER
Earth-fault withstand	≥ 20 kA for 3 s (or ≥ 25 kA for 1 s) per EN 60076-5
MV voltage	33 kV (highest equipment 36 kV); insulation 170 BIL / 70 AC
Auxiliary winding	≥ 315 kVA @ 400/230 V, 3-ph 4-wire (minimum — upratable; quote 315/400/500 kVA)
Cooling / oil	ONAN; mineral oil EN 60296
Standards	EN/IEC 60076-1/-2/-3/-5; IEC 60076-6; EN 60289
Note	Full requirements in the attached T2 datasheet (Excel)

2. PROGRAMME — Q1 2027 DELIVERY

2.1 Target manufacturing slot: January 2027.

2.2 Target on-site delivery: Q1 2027 (DAP Limassol Port / Plot 26 Psevdas), aligned with the Client's implementation programme and TSOC connection schedule. State your manufacturing lead time and total delivery time to Limassol Port against this target.

3. COMPLIANCE & OPEN ITEMS

3.1 Return a signed clause-by-clause compliance statement (COMPLY / DEVIATE + note). Explicitly confirm: YNd11 (not Dyn11), uk 21%, 170/70 LV insulation, Tier 2 Ecodesign PEI, and that T2 is included. State no-load loss, load loss and PEI % for both units.

3.2 Bid the following open items as instructed (do not assume):

- T2 auxiliary winding: bid 315 kVA minimum, confirm upratable, give price per kVA step (315/400/500).
- T1 132 kV CTs: quote 6-CT base PLUS a priced option for 9-CT (neutral CT for REF) — final fixed when the ISM bay protection single-line is issued.
- Seismic: design to CYS EN 1998-1 Zone II, agR 0.23 g, importance factor 1.4.

4. DELIVERY & INCOTERMS

- State ex-works manufacturing lead time (weeks from order/LOI).
- State total delivery time to Limassol Port, Cyprus (incl. transit + customs clearance).
- Price DAP Limassol Port (Incoterms 2020) as baseline; also quote DAP/DDP Plot 26 Psevdas (inland haulage + offloading).
- Transport provisions: impact recorders w/ GPS, lifting/jacking points, shipping oil/nitrogen state, abnormal-load responsibility, any port-handling exclusions.

5. CERTIFICATIONS (attach with bid)

- EU Declaration of Conformity + CE marking.
- EU Ecodesign Tier 2 (Reg. 548/2014 as amended by 2019/1783) — PEI declaration.
- Independent short-circuit withstand TYPE TEST to EN 60076-5 (KEMA / CESI / IPH) — calculation-only not accepted.
- Temperature-rise (EN 60076-2); impulse & AC withstand (EN 60076-3) at the 170/70 LV level; sound level (EN 60076-10).
- Routine test certificates per EN 60076-1; material certs (oil EN 60296, bushings EN 60137, OLTC EN 60214).
- ISO 9001 / 14001 / 45001; EU references for 132 kV / ≥63 MVA units; warranty & LTSA terms.

6. COMMERCIAL & SUBMISSION

- Prices in EUR, itemised: T1, T2, transport/Incoterm options, spares, supervision of erection & commissioning.
- Offer validity: minimum 90 days.
- Warranty: 24 months from commissioning or 36 months from delivery, whichever comes first (60 months from delivery preferred). Must cover BOTH T1 and T2. State terms explicitly.
- Payment terms: state proposed (final per executed EPC).
- Submit to office@lighthief.com by the bid deadline: Tuesday 17 June 2026, 18:00 (Cyprus / EEST).
- Format: completed T1 questionnaire (OFFER column) + completed T2 datasheet + compliance statement + certificates + commercial offer.

WHY THIS DEADLINE: Bids must arrive by 17 June 2026 so Lighthief can evaluate all offers and present the consolidated solution to the Client at Intersolar Europe (23–25 June 2026). Late or incomplete bids may be excluded.

7. EVALUATION CRITERIA

- Technical compliance (T1 and T2), incl. genuine Tier 2 Ecodesign & independent short-circuit type test.

- Total cost of ownership (price + capitalised losses / PEI).
- Delivery time to Limassol vs January 2027 slot / Q1 2027 target.
- Manufacturer track record & EU references; warranty / LTSA.

Contact for clarifications: Alexander Papacosta, Cyprus Director — office@lighthief.com — +357 99 164 158 — solarfarms.cy.

Lighthief Cyprus Ltd | HE 477423 | 28 October Ave 249, Lophitis Business Center 1, Office 201, 3035 Limassol, Cyprus
office@lighthief.com | +357 77 77 00 50 | solarfarms.cy | Ref: LCY-RFP-HESS-TRF-2026